CONCEPTS IN BIG DATA ANALYTICS

COMPREHENSIVE STUDY GUIDE

📚 MODULE 1: INTRODUCTION TO BIG DATA (9 hrs)

1.1 What is Big Data?

- Definition and basic concepts
- Differences between conventional data and big data
- Evolution of Big Data
- ***Exam Reference:*** 2023 Q1, 2024 Q12(a)

1.2 Big Data Architecture

- Components and layers
- Architectural framework with diagrams
- ***Exam Reference:*** 2024 Q11(a)

1.3 Big Data Platforms

- Major platforms overview
- Features and capabilities comparison
- ***Exam Reference:*** 2023 Q2, 2024 Q2

1.4 Types of Data in Big Data

- Structured, unstructured, and semi-structured data
- Examples of each type
- Nature of data and its applications
- ***Exam Reference:*** 2024 Q12(b)

1.5 Data Analytic Processes and Tools

- Key processes in Big Data analytics
- Common tools used in the industry
- ***Exam Reference:*** 2023 Q12(b)

1.6 The 5 V's of Big Data

- Volume, Velocity, Variety, Veracity, Value

- Examples for each characteristic
- ***Exam Reference:*** 2023 Q12(a)

1.7 Big Data Analytical Methods

- Descriptive, diagnostic, predictive, and prescriptive analytics
- Applications and use cases
- Data analysis vs. data reporting
- ***Exam Reference:*** 2023 Q11(a), 2024 Q1

1.8 Intelligent Data Analysis

- Methods and approaches
- Comparison with traditional analysis
- ***Exam Reference:*** 2023 Q11(b)

1.9 Big Data Analytics Life Cycle

- Stages and processes
- Implementation considerations
- ***Exam Reference:*** 2024 Q11(b)

📚 MODULE 2: INTRODUCTION TO STREAM COMPUTING (8 hrs)

2.1 Stream Computing Fundamentals

- Definition and concepts
- Comparison with batch processing
- ***Exam Reference:*** 2023 Q4

2.2 Streaming Data Architecture

- Components and design
- Implementation considerations
- ***Exam Reference:*** 2023 Q3

2.3 Stream Data Models

- Characteristics and structure
- Differences from traditional databases

2.4 Sampling in Stream Processing

- Importance and applications

- Various sampling techniques
- ***Exam Reference:*** 2024 Q14(a)

2.5 Bloom Filters

- Concept and properties
- Working mechanism with examples
- Operations supported by Bloom filters
- ***Exam Reference:*** 2023 Q13(a), 2024 Q13(a)

2.6 Flajolet-Martin Algorithm

- Purpose and implementation
- Applications in stream data processing
- Finding distinct elements in a stream
- ***Exam Reference:*** 2023 Q13(b), 2024 Q13(b)

2.7 Moments Estimation in Data Streams

- Concept and importance
- Types of moments
- ***Exam Reference:*** 2023 Q14(b), 2024 Q4

2.8 DGIM Algorithm

- Purpose and implementation
- Rules for forming buckets
- Applications in stream computing
- Window-based counting
- ***Exam Reference:*** 2023 Q14(a), 2024 Q14(b)

2.9 Challenges in Stream Processing

- Technical and implementation challenges
- Solutions and approaches
- ***Exam Reference:*** 2024 Q3

📚 MODULE 3: HADOOP DISTRIBUTED FILE SYSTEM (13 hrs)

3.1 History and Development of Hadoop

- Origins and evolution

- Need for Hadoop in Big Data Analytics
- ***Exam Reference:*** 2023 Q5

3.2 Hadoop Ecosystem

- Core components
- Interaction between components
- ***Exam Reference:*** 2024 Q16(b)

3.3 HDFS Architecture

- Design philosophy
- Components and their functions
- Namenodes vs. Datanodes
- ***Exam Reference:*** 2024 Q5

3.4 HDFS File Storage and Management

- Block storage mechanism
- Replication strategy
- File systems in Hadoop
- ***Exam Reference:*** 2023 Q6

3.5 HDFS Operations

- Reading files in HDFS
- Writing files in HDFS
- Basic file system commands
- ***Exam Reference:*** 2023 Q15(b), 2024 Q15(b)

3.6 MapReduce Framework

- Concept and implementation
- Execution pipeline
- Job execution flow
- ***Exam Reference:*** 2023 Q16(a), 2024 Q15(a)

3.7 Parallel Processing with MapReduce

- How MapReduce enables parallelism
- Runtime coordination and task management
- ***Exam Reference:*** 2023 Q16(b), 2024 Q16(a)

3.8 MapReduce Examples

- WordCount implementation
- Road Enrichment example
- ***Exam Reference:*** 2023 Q15(a), 2024 Q6

📚 MODULE 4: PIG, HIVE, AND HBASE (7 hrs)

4.1 Apache Pig

- Introduction and purpose
- Architecture and components
- ***Exam Reference:*** 2024 Q17(a)

4.2 Pig Execution Modes

- Local mode
- MapReduce mode
- ***Exam Reference:*** 2023 Q7

4.3 Pig vs. Traditional Databases

- Comparison with SQL
- Advantages and limitations
- ***Exam Reference:*** 2023 Q8, 2024 Q7

4.4 Grunt and Pig Latin

- Grunt shell usage
- Pig Latin syntax and features
- Data types and operators
- ***Exam Reference:*** 2024 Q18(b)

4.5 User-Defined Functions in Pig

- Purpose and implementation
- Examples of UDFs

4.6 Hive Architecture

- Components and design
- Services provided by Hive
- ***Exam Reference:*** 2024 Q17(b)

4.7 Hive Metastore

- Function and importance
- Comparison with traditional databases

4.8 HiveQL and Tables

- Query language features
- Table creation and partitioning
- Types of tables in Hive
- ***Exam Reference:*** 2023 Q17(b), 2024 Q18(a)

4.9 HBase

- Concepts and purpose
- Architecture and features
- Differences from RDBMS
- Data model and operations
- ***Exam Reference:*** 2023 Q18(b), 2024 Q8

📚 MODULE 5: INTRODUCTION TO R PROGRAMMING (8 hrs)

5.1 R Programming Fundamentals

- Introduction and importance
- Key features and advantages
- ***Exam Reference:*** 2023 Q9, 2024 Q9

5.2 R Graphical User Interfaces

- Types and features
- Usage scenarios

5.3 Vectors in R

- Creation and manipulation
- Operations on vectors
- ***Exam Reference:*** 2024 Q19(a)

5.4 Matrices in R

- Creation methods
- Matrix operations

- Applying functions to rows and columns
- ***Exam Reference:*** 2023 Q20(a)

5.5 Lists in R

- Definition and usage
- Difference from vectors and matrices
- List operations
- ***Exam Reference:*** 2024 Q20(a)

5.6 Data Frames in R

- Concept and creation
- Comparison with matrices
- Operations on data frames
- ***Exam Reference:*** 2023 Q19(a), 2024 Q20(b)

5.7 File Operations in R

- Reading data
- Writing data
- ***Exam Reference:*** 2023 Q19(b)

5.8 R Programming Examples

- Factorial program
- Fibonacci series program
- Dataset analysis
- ***Exam Reference:*** 2023 Q10, 2024 Q10

5.9 Data Types and Attributes in R

- Categories and classifications
- Usage in data analysis
- ***Exam Reference:*** 2024 Q19(b)

📝 EXAM PREPARATION TIPS

1. **Focus on frequently asked topics:**
    - Bloom filters and stream algorithms
    - HDFS operations
    - MapReduce framework

- R programming applications

2. **Practice programming questions:**
   - R programs for factorial, Fibonacci series
   - MapReduce implementations
   - Pig Latin and HiveQL queries
3. **Master diagram-based questions:**
   - Big Data architecture
   - HDFS architecture
   - Hadoop ecosystem
   - Hive and Pig architecture
4. **Understand algorithms with examples:**
   - DGIM algorithm
   - Flajolet-Martin algorithm
   - Bloom filter operations
5. **Prepare short answers for Part A questions:**
   - Differences between concepts
   - Features and characteristics
   - Basic definitions and comparisons

This study guide is arranged for optimal comprehension and exam preparation. Use it alongside course materials for best results.